Assignment 1

Problem 1

Assume that reviews of Movie i have a common probability p_i of being positive and independent. Assume $U(0,1)$ prior on each p_i.

	Positive	Not Positive	Total
Movie 1	425	75	500
Movie 2	9	1	10

1. Determine the posterior distribution of p_1 and p_2 separately.

1. Priors:

$$\begin{aligned} &\because p_i \sim U(0,1) \\ &\therefore p_1 \sim U(0,1) \quad p_2 \sim U(0,1) \end{aligned}$$

2. Sampling Distribution

- As the reviews are either `positive` or `not positive`, so the sampling distribution for each p_i is `Binomial`.
- $p(y = 425 \mid p_1) \sim Bin(500, p_1)$

$$\begin{aligned} p(y = 425 \mid p_1) &= \binom{500}{425} p_1^{425}(1-p_1)^{75} \\ &\propto p_1^{425}(1-p_1)^{75} \end{aligned}$$

- $p(y = 9 \mid p_2) \sim Bin(10, p_2)$

$$\begin{aligned} p(y = 9 \mid p_2) &= \binom{10}{9} p_2^{9}(1-p_2) \\ &\propto p_2^{9}(1-p_2) \end{aligned}$$

3. Posterior Distribution

- $p(p_1|y = 425)$

$$\begin{aligned} &p(p_1 \mid y = 425) \propto 1 \cdot p_1^{425}(1-p_1)^{75} \\ &\implies p_1 \mid y = 425 \sim Beta(426, 76) \end{aligned}$$

- $p(p_2|y = 9)$

$$\begin{aligned} &p(p_2 \mid y = 9) \propto 1 \cdot p_2^{9}(1-p_2) \\ &\implies p_2 \mid y = 9 \sim Beta(10, 2) \end{aligned}$$

- Answers

$$\boxed{p_1 \mid y = 425 \sim Beta(426, 76) \quad p_2 \mid y = 9 \sim Beta(10, 2)}$$

2. Which movie ranks higher?

```r
# parameters
p1.alpha <- 426
p1.beta <- 76
p2.alpha <- 10
p2.beta <- 2

# summary function
beta_summary <- function(alpha, beta) {
    data.frame(
        mean = alpha / (alpha + beta),
        mode = (alpha - 1) / (alpha + beta - 2),
        median = qbeta(0.5, alpha, beta)
    )
}

# generate summary
p1 <- beta_summary(p1.alpha, p1.beta)
p2 <- beta_summary(p2.alpha, p2.beta)
higher_rank <- data.frame(
    mean = ifelse(p1$mean > p2$mean, "p1", "p2"),
    mode = ifelse(p1$mode > p2$mode, "p1", "p2"),
    median = ifelse(p1$median > p2$median, "p1", "p2")
)
result <- rbind(p1, p2, higher_rank)
rownames(result) = c("p1", "p2", "higher rank")
result
```

```
##                          mean mode            median
## p1          0.848605577689243 0.85 0.849068693158005
## p2          0.833333333333333  0.9 0.852036574569394
## higher rank                p1   p2                p2
```

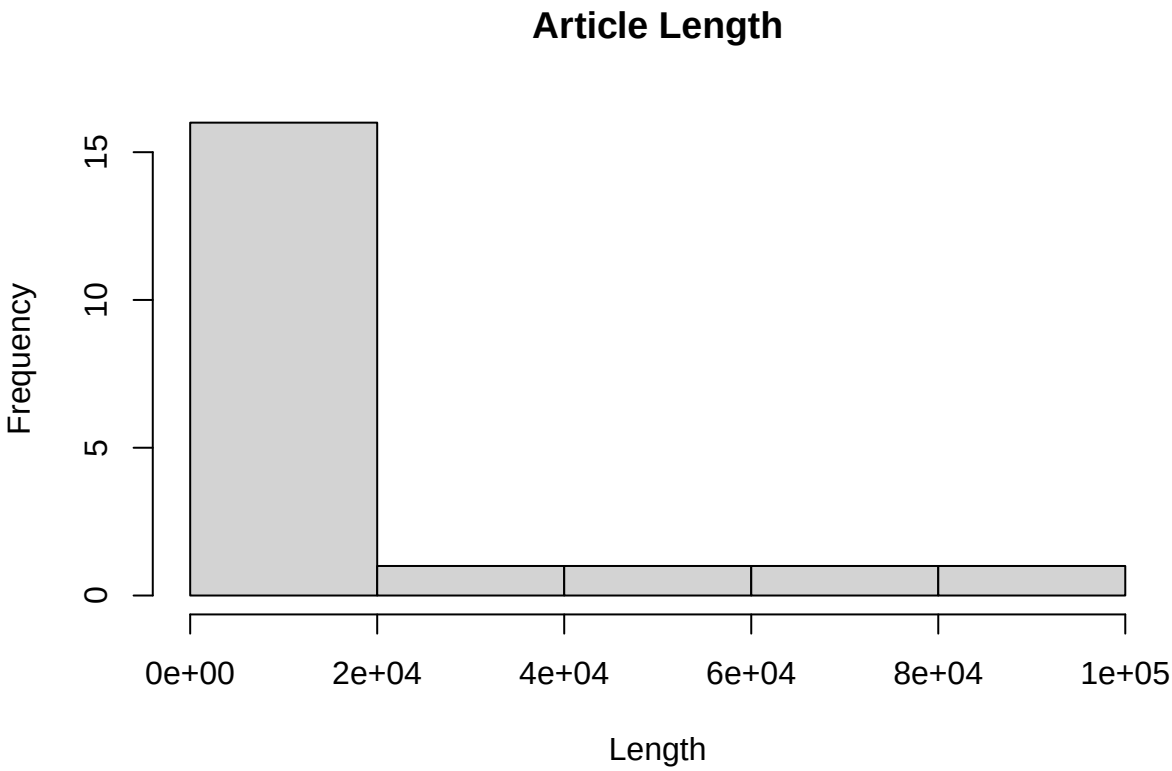
Problem 2

a. Complete the following using R

a.i. Display a histogram

- Given the following histogram and summary, we can know that the distribution is strongly right-skewed ($mean > median$).

```
data <- read.table("./Random Wikipedia.txt", header = TRUE)
hist(data$bytes, xlab = "Length", main = "Article Length")
```




```
summary(data$bytes)
```

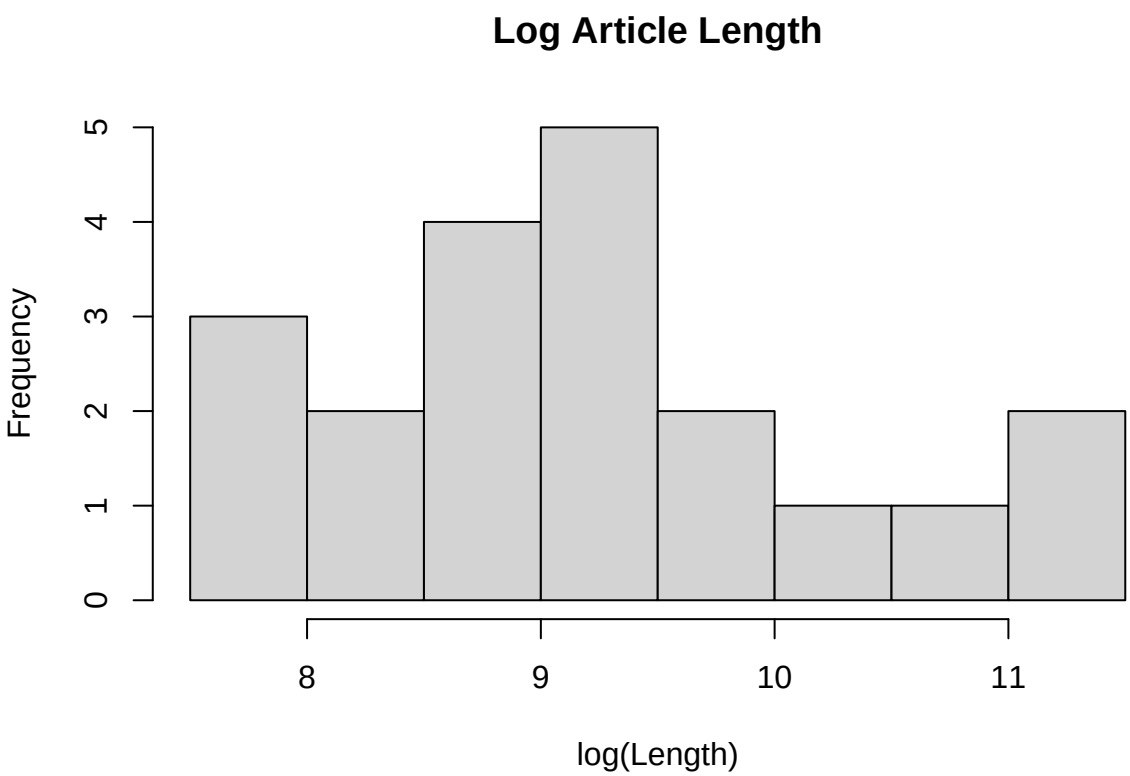
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2132    4982   10294   18827   16806   86979
```

---

a.ii. Transforming article length to the log scale

- After rescaling the distribution with natural log, the distribution is more symmetrical now ($mean \approx median$)

```
log_length <- log(data$bytes)
hist(log_length, xlab = "log(Length)", main = "Log Article Length")
```




```r
summary(log_length)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   7.665   8.512   9.238   9.252   9.727  11.373
```

---

a.iii. Transforming article length to the log scale

- Log scale improves the reliability of the mean value as it reduces the impact of the outliers on the mean value.

---

b. Compute the sample mean and sample variance

```r
n <- length(log_length)
y_bar <- mean(log_length)
s.2 <- var(log_length)

results <- data.frame(mean = y_bar, variance=s.2)
colnames(results) = c("sample mean", "sample variance")
results
```

```
##   sample mean sample variance
## 1     9.25151        1.138061
```

---

c. Assume $y \sim N(\mu, \sigma^2)$, $sigma^2 = s^2$ and the prior is flat for μ

c.i. Posterior Mean, Posterior Variance, and Posterior Precision

Prior

- Given that prior is flat,

$$p(\mu) \propto 1 \quad (\mu \in (-\infty, \infty))$$

Sampling Distribution

- Let random variables $y_1, \ldots, y_{20}$ be a random vector y

$$\begin{aligned}
&y \mid \mu, \sigma^2 \sim N(\mu, \sigma^2) \\
&p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right) \\
&p(y \mid \mu, \sigma^2) \propto \frac{1}{\sigma} \exp\left(-\frac{n-1}{2\sigma^2}s^2\right) \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{y})^2\right) \quad \text{(Mean only)} \\
&p(y \mid \mu) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{y})^2\right)
\end{aligned}$$

Posterior

$$\begin{aligned}
p(\mu \mid y) &\propto p(\mu)p(y \mid \mu) \\
&\propto 1 \cdot \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{y})^2\right) \\
&\propto \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{y})^2\right) \\
&\implies \boxed{\mu \mid y \sim N(\bar{y}, \sigma^2/n)}
\end{aligned}$$

```r
# posterior mean
posteriors <- data.frame(
    mu_mean = y_bar,
    mu_variance = s.2 / n,
    mu_precision = n / s.2
)
posteriors
```

```
##   mu_mean mu_variance mu_precision
## 1 9.25151  0.05690307     17.57374
```

---

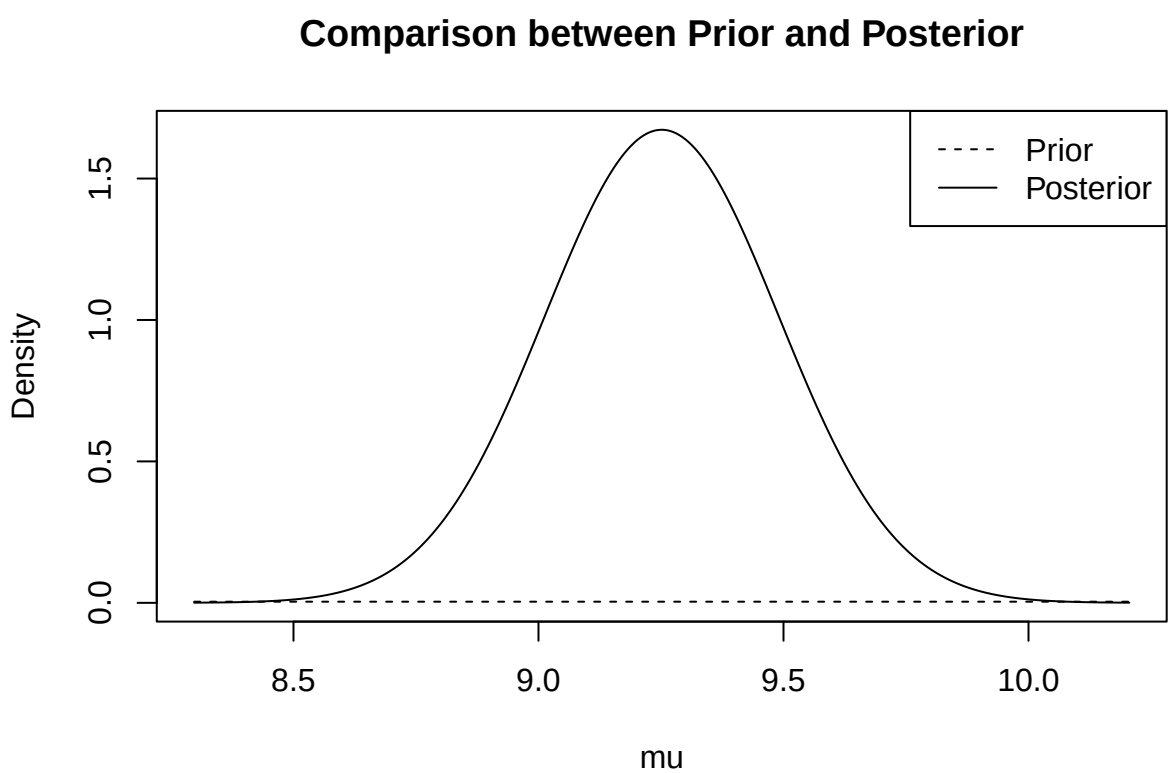
c.ii. Comparison between Prior μ and Posterior $\mu \mid y$

```r
# Posterior density
post_std <-sqrt(posteriors$mu_variance)
curve(
    dnorm(x, posteriors$mu_mean, post_std),
    from = posteriors$mu_mean - 4 * post_std,
    to = posteriors$mu_mean + 4 * post_std,
```

```r
    xlab = "mu",
    ylab = "Density",
    main = "Comparison between Prior and Posterior",
    n = 1000
)

# Prior density
# large variance = 10000 to make prior diffuse -> flat prior
curve(
  dnorm(x, y_bar, 100),
  lty = 2, n = 1000, add = TRUE
)

legend("topright", c("Prior", "Posterior"), lty = c(2, 1))
```



c.iii. 95% central posterior interval for μ

```r
qnorm(c(0.025, 0.975), mean=posteriors$mu_mean, sd=post_std)
```

```
## [1] 8.783973 9.719047
```

---

d. Treat $sigma^2$ as unknown now, and let μ and σ^2 have prior $p(\mu, \sigma^2) \propto (sigma^2)^{-1}$

d.i. Posterior Mean, Posterior Variance, and Posterior Precision

Likelihood from

$$p(y \mid \mu, \sigma^2) \propto \frac{1}{\sigma} \exp\left(-\frac{n-1}{2\sigma^2} s^2\right) \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{y})^2\right)$$

Posterior

$$
\begin{aligned}
p(\mu \mid y) &\propto p(y \mid \mu, \sigma^2) p(\mu, \sigma^2) \\
&\propto \frac{1}{\sigma} \exp\left(-\frac{n-1}{2\sigma^2} s^2\right) \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{y})^2\right) \cdot (\sigma^2)^{-1} \\
&\implies \boxed{\mu \mid \sigma^2, y \sim N(\bar{y}, \sigma^2/n)} \\
&\implies \boxed{\sigma^2 \mid y \sim Inv - \chi^2(n-1, s^2)}
\end{aligned}
$$

Posterior Mean, Posterior Variance, Posterior Precision

- We simulate σ^2 by sampling X from χ^2 distribution and rewrite the σ^2 distribution as:

$$
\sigma^2_{sim} = \frac{(n-1)s^2}{X} \quad X \sim \chi^2_{n-1}
$$

```r
# simulation
sigma_sim.2 <- (n - 1) * s.2 / rchisq(1000000, n - 1)
mu_sim <- rnorm(1000000, y_bar, sqrt(sigma_sim.2 / n))

posteriors <- data.frame(
    mu_mean = mean(mu_sim),
    mu_variance = var(mu_sim),
    mu_precision = 1 / var(mu_sim)
)
posteriors
```

```
##    mu_mean mu_variance mu_precision
## 1 9.251747  0.06349026     15.75045
```

---

d.ii. 95% central posterior interval for μ

```r
quantile(mu_sim, c(0.025,0.975))
```

```
##     2.5%    97.5%
## 8.752310 9.750229
```

---

d.ii. 95% central posterior interval for σ^2

```r
quantile(sigma_sim.2, c(0.025,0.975))
```

```
##      2.5%     97.5%
## 0.6586988 2.4275660
```

---

e. 1,000,000 draws from the posterior

```r
N = 1000000
sigma_sim.2 <- (n - 1) * s.2 / rchisq(N, n - 1)
mu_sim <- rnorm(N, y_bar, sqrt(sigma_sim.2 / n))
post.pred <- exp(rnorm(N, mu_sim, sqrt(sigma_sim.2)))
```

e.i. 95% central posterior predictive interval for the length

```r
quantile(post.pred, c(0.025, 0.975))
```

```
##       2.5%      97.5%
##   1058.922 103543.235
```

---

e.ii. The posterior predictive probability that min length of a newly selected articles will be less than the minimum article length in the original data

```r
min_length <- min(data$bytes)
mean(post.pred < min_length)
```

```
## [1] 0.08163
```

---

e.iii. The posterior predictive probability that min length of 20 newly selected articles will be less than the minimum article length in the data

```r
records <- logical(N)
for (i in seq_len(N)) {
    cur_samples <- exp(rnorm(20, mu_sim[i], sqrt(sigma_sim.2[i])))
    records[i] <- min(cur_samples) < min_length
}
mean(records)
```

```
## [1] 0.735303
```